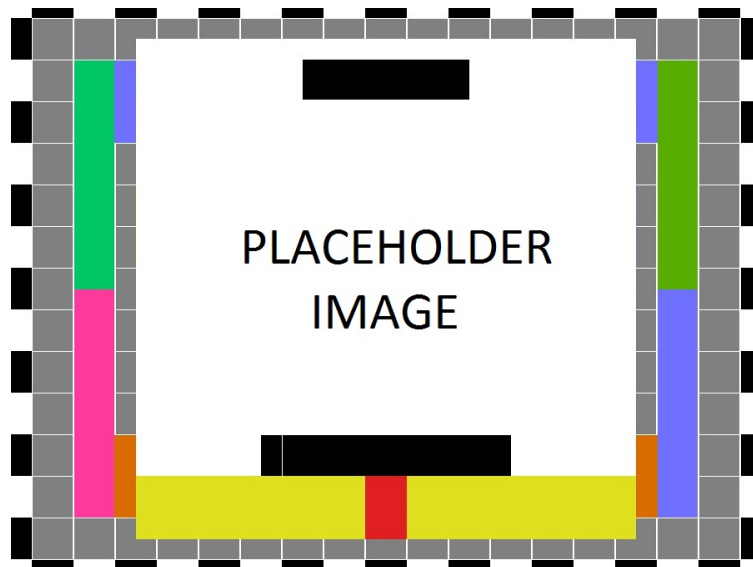


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**ARCHITECTURAL DESIGN SPECIFICATION  
CSE 4316: SENIOR DESIGN I  
FALL 2026**



**TEAM PERCEPTRON  
DIGITDESK HANDWRITTEN FORM DIGITIZATION  
SERVICE**

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## REVISION HISTORY

| Revision | Date       | Author(s)  | Description       |
|----------|------------|------------|-------------------|
| 0.1      | 11.02.2026 | JVN        | document creation |
| 1.0      | 11.18.2026 | AT, GH, CB | official release  |

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1 INTRODUCTION

Architecture of DigitDesk Handwritten Form Digitization Service as an ISO/IEC/IEEE 42010:2022 [1] architecture description using IEEE 1016-2009 [2] viewpoints. Refines the SRS. Scope: the on-prem service, its REST API, and the review UI.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

| Stakeholder              | Primary concerns                                    |
|--------------------------|-----------------------------------------------------|
| Sponsor (records office) | accuracy, auditability, on-prem data handling, cost |
| Clerks (reviewers)       | fast, clear review workflow                         |
| IT administrator         | deployability, monitoring, security                 |
| Development team         | feasibility, modular/replaceable model, testability |

3 ARCHITECTURE VIEWPOINTS

Context (Section 4), Composition/Data Flow (Section 5), Logical/Functional (Sections 6–8), Deployment (Section 8). Style: a containerized pipeline of services communicating over an internal message queue and HTTP, deployed on-prem. Interfaces use #NN; subsystems named [Layer] . [Name].

4 SYSTEM OVERVIEW & CONTEXT VIEW

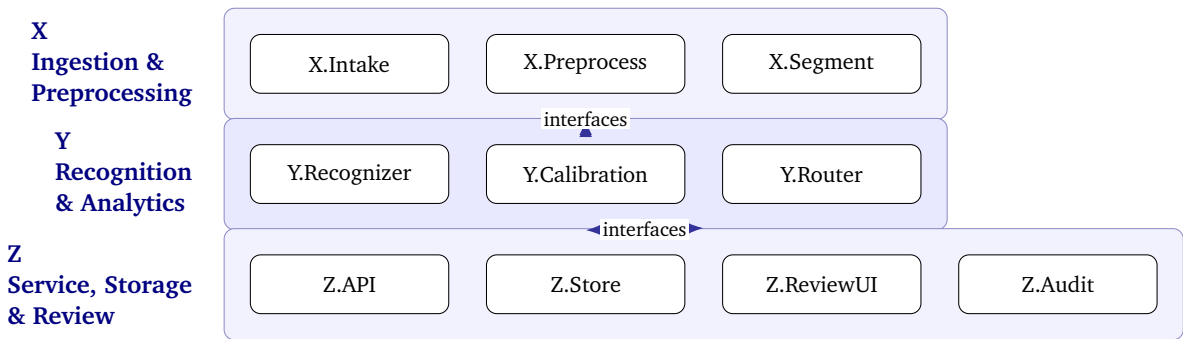


Figure 1: Three-layer pipeline: ingestion, recognition, service

External actors: the scan workflow (submits images to the API), the clerk (review UI), and the downstream records database (consumes verified export). X ingests and prepares images; Y recognizes and routes; Z serves, stores, reviews, and audits.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

6 X LAYER SUBSYSTEMS — INGESTION & PREPROCESSING

6.1 X.INTAKE

**Responsibilities:** accept uploads (EI-01), assign a job id, enqueue work.

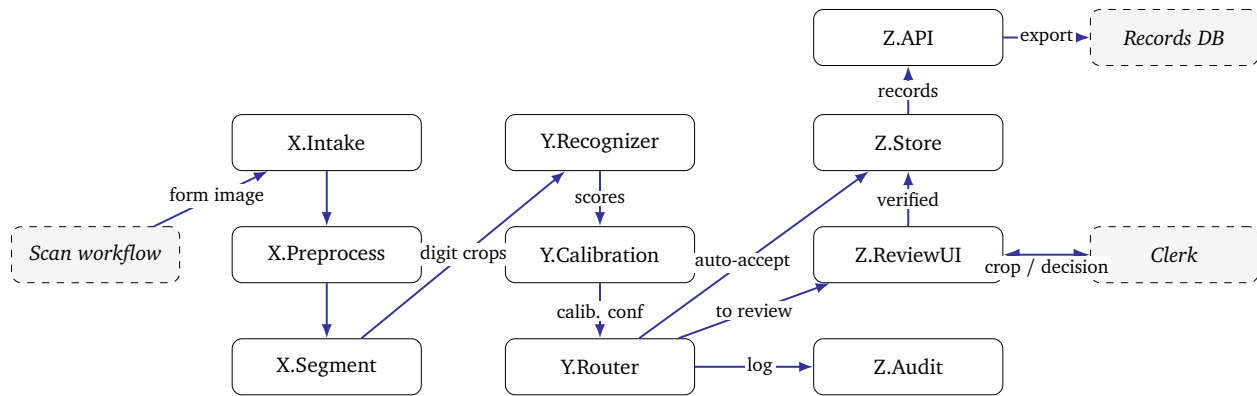


Figure 2: Image → preprocess → segment → recognize → route → store/review

| ID  | Description   | Inputs     | Outputs              |
|-----|---------------|------------|----------------------|
| #01 | Ingestion API | form image | job id, queued image |

## 6.2 X.PREPROCESS

**Responsibilities:** deskew, denoise, binarize to a normalized page.

## 6.3 X.SEGMENT

**Responsibilities:** locate numeric fields on the known template and emit per-field digit crops (FR-01).

**Rationale:** template-anchored segmentation is robust and simple for a fixed form (AD-01).

# 7 Y LAYER SUBSYSTEMS — RECOGNITION & ANALYTICS

## 7.1 Y.RECOGNIZER

**Responsibilities:** classify digit crops and output values with raw scores (FR-02). Model is a replaceable artifact (QA-01).

| ID  | Description | Inputs      | Outputs            |
|-----|-------------|-------------|--------------------|
| #02 | Recognition | digit crops | values, raw scores |

## 7.2 Y.CALIBRATION

**Responsibilities:** convert raw scores to calibrated confidences (temperature scaling), so routing thresholds are meaningful (supports PE-02, FR-03).

## 7.3 Y.ROUTER

**Responsibilities:** apply the acceptance threshold (FR-03) — auto-accept or enqueue for review.

# 8 Z LAYER SUBSYSTEMS — SERVICE, STORAGE & REVIEW

## 8.1 Z.API

**Responsibilities:** REST surface, authentication (SE-02), job status, export (FR-05). **Deployment:** containers orchestrated on the on-prem host.

## 8.2 Z.STORE

**Responsibilities:** persist images, results, and versions (MS-01); on-prem only (SE-01).

## 8.3 Z.REVIEWUI

**Responsibilities:** present queued fields for confirm/correct (FR-04).

## 8.4 Z.AUDIT

**Responsibilities:** append-only log of every field decision (FR-05, SE-02).

# 9 ARCHITECTURE DECISIONS & RATIONALE

## 9.1 AD-01: TEMPLATE-ANCHORED SEGMENTATION

**Context:** fixed form layout (FR-01). **Decision:** anchor fields to the template rather than learn detection. **Alternatives:** learned detector (more data, more risk). **Consequences:** simple and robust now, less general (see FI-01). **Affects:** FR-01.

## 9.2 AD-02: CALIBRATED CONFIDENCE FOR ROUTING

**Context:** FR-03/PE-02 depend on trustworthy confidence. **Decision:** add a calibration stage. **Alternatives:** raw softmax (poorly calibrated). **Consequences:** extra stage, but auto-accept error stays bounded. **Affects:** FR-02, FR-03, PE-02.

# 10 REQUIREMENTS TRACEABILITY

| Req ID | Requirement (abbrev.)    | Realizing subsystem(s)      |
|--------|--------------------------|-----------------------------|
| FR-01  | Field segmentation       | X.Segment                   |
| FR-02  | Recognition + confidence | Y.Recognizer, Y.Calibration |
| FR-03  | Confidence routing       | Y.Router                    |
| FR-04  | Human review             | Z.ReviewUI                  |
| FR-05  | Export & audit           | Z.API, Z.Audit              |
| SE-01  | On-prem data             | Z.Store                     |

# REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.